

Critical Literature Review

What Each Source Adds, Inspires, or Debates for a Monthly Study of
Thermal Extremes and Cardiovascular Admissions in Hong Kong, 2013–2023

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Quality over quantity. This note is not a catalogue of every temperature–health paper. It is a working map from a small set of high-leverage sources to concrete choices in research design, modelling, and the project’s R pipeline.

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1 Purpose and reading guide

The project asks whether *officially defined* thermal extremes—especially hot nights and cold days—are associated with monthly principal-diagnosis inpatient admissions for acute myocardial infarction (AMI), ischemic stroke, and hemorrhagic stroke among Hong Kong residents aged 35+, during 2013–2023, in an aging city that still experiences cold.

That question sits inside a crowded Hong Kong literature. Claiming “first” is unsafe. Claiming “nothing is known” is false. The useful task is sharper: identify what prior work already settles, what it leaves open, and what it forces us to change in our own design and code.

Each core source below is read through four lenses:

- **Adds.** Empirical or conceptual content we should treat as established background.
- **Inspires.** A design idea, metric, or modelling habit worth adopting or adapting.
- **Debates.** A tension with our current plan, a novelty constraint, or a claim we must weaken.
- **Implementation.** Concrete implications for `config.yml`, scripts, data requests, or interpretation rules.

A recurring intellectual distinction runs through the review:

Trigger versus burden. Daily distributed-lag models estimate acute *triggering* of events over specific lags. Monthly extreme-count models estimate broader hospital-*burden* associations. Both are scientifically legitimate; they answer different questions and produce non-comparable coefficients (Bhaskaran et al., 2013; Gasparrini et al., 2010).

2 Hong Kong historical baseline: cold-dominant cardiovascular risk

2.1 Goggins et al. (2013) — AMI, weather, and pollution

Adds. In Hong Kong (2000–2009), daily AMI admissions rose as temperatures cooled below an estimated $\sim 24^{\circ}\text{C}$ threshold ($\sim 3.7\%$ per 1°C using multi-day lags). A statistically significant heat effect was not the

main story in that decade. Same-day NO_2 was the strongest pollutant predictor among those examined (Goggins et al., 2013).

Inspires. Keep cold as a *co-primary* exposure, not a historical footnote. Treat heat as an open empirical question rather than a presumed replacement for cold. Retain NO_2 in staged pollution models for AMI.

Debates. This paper cannot be used as a numeric benchmark for our monthly rate ratios. Different decade, daily mean temperature, and lag structure. A “regime shift from cold to heat” claim would require methodologically harmonized comparisons that we do not currently have.

Implementation. Primary model should include cold days alongside hot nights (`scripts/11_*.R` already does this). Do not enter highly collinear heat metrics together. Avoid direct numeric comparison tables against Goggins per- $^\circ\text{C}$ estimates in manuscripts.

2.2 Goggins et al. (2012) — stroke subtypes separately

Adds. Hemorrhagic stroke (HS) showed a strong, approximately linear inverse association with mean temperature across the observed range (lags 0–4; $\sim 2.7\%$ higher admissions per 1°C lower). Ischemic stroke (IS) was weaker and mainly apparent below $\sim 22^\circ\text{C}$ with longer lags (0–13). Pollutants were not associated with either subtype. Effects were stronger among older adults and women for HS (Goggins et al., 2012).

Inspires. Never pool “stroke” as the primary endpoint. Model AMI, IS, and HS separately. Expect different lag biology even if our monthly design cannot resolve daily lags: HS may be more acute; IS more delayed/cold-threshold.

Debates. Monthly aggregation may attenuate short-lag HS signals more than longer-lag IS/cold signals. A null monthly HS–heat association would not refute daily HS–cold biology.

Implementation. Keep diagnosis-stratified models mandatory (already planned). Prefer principal diagnosis. Request sex-specific extracts. Age bands should separate 65–69 and 70–74 rather than collapsing into “65+”.

2.3 Lam et al. (2018) — diabetes as effect modifier for AMI

Adds. Among public-hospital AMI admissions (2002–2011), patients with diabetes showed stronger cold-season temperature associations than patients without diabetes, and some hot-season sensitivity that non-DM patients lacked (Lam et al., 2018).

Inspires. Vulnerability is not only chronological age. Comorbidity can reshape the temperature–AMI curve.

Debates. Aggregate monthly extracts may not support diabetes stratification. If comorbidity is unavailable, we must not pretend age alone captures the Lam finding.

Implementation. Ask Roro whether a diabetes flag (or DM-related secondary diagnosis) is feasible as a sensitivity stratum. If not, document this as a known limitation in the assumption ledger (A27) rather than inventing proxies.

3 The decisive recent overlap: hot nights are not a blank slate

3.1 Guo et al. (2024) — official hot nights versus nighttime heat intensity

This is the single most important paper for our novelty and exposure strategy.

Adds. Using daily Hong Kong hospitalizations in hot seasons (May–October, 2000–2019), Guo et al. compared three nighttime metrics (Guo et al., 2024):

- **HNday28°C**: official hot night ($T_{\min} \geq 28^\circ\text{C}$);
- **HNe**: hot-night excess (hourly degrees above 28°C at night, $^\circ\text{C}\cdot\text{h}$);
- **HNday90th**: nights above the 90th percentile of HNe.

Extreme HNe was associated with higher non-cancer non-external hospitalizations (about 3.1% over lags 0–4), with larger effects in older and lower-SES groups and for circulatory admissions. Critically, the official binary HNday28°C metric did *not* show an overall excess-risk association, whereas intensity-based metrics did.

Inspires. Distinguish *policy metrics* from *biological metrics*. Official binary hot nights remain policy-aligned and communicable. Intensity and spell metrics may be closer to physiological stress. Consecutive intensive hot nights mattered; late-season unusual nighttime heat was especially hazardous in their interpretation.

Debates. We cannot claim to be the first Hong Kong hot-night hospitalization study. We also cannot treat a null finding for monthly official hot-night counts as proof that nighttime heat is harmless. Guo’s result predicts that binary counts may be weak, while intensity/spell alternatives may be more informative.

Implementation.

1. Keep official monthly hot-night counts as the *primary policy-aligned* heat exposure.
2. Prespecify intensity/spell alternatives as co-equal scientific checks, not post-hoc rescue analyses.
3. Request hourly temperature (or precomputed HNe) from Hogan if feasible; daily $T_{\min}$ alone cannot reconstruct true HNe.
4. Until hourly data arrive, implement daily-derived spell summaries already foreshadowed in `scripts/02.build_weather` (longest run; days in spells ≥ 3 or ≥ 5 ; 2D3N-style event flags).
5. In manuscripts, state explicitly: “A null official-hot-night association would be compatible with Guo et al. (2024) and would not close the nighttime-heat question.”

3.2 Wang et al. (2019) — extremely hot weather events as combinations

Adds. Using official VHD ($T_{\max} \geq 33^\circ\text{C}$) and HN ($T_{\min} \geq 28^\circ\text{C}$) definitions, Wang et al. showed that event *structure* matters for mortality: single hot nights raised risk; five or more consecutive VHDs/HNs raised it further; combined patterns such as 2D3N (two very hot days with three hot nights) were especially informative; females and older adults were more vulnerable (Wang et al., 2019).

Inspires. Count-based monthly totals discard sequence. Two months with ten hot nights can differ radically if one has scattered nights and the other has a five-night siege. Spell and combination metrics recover some of that information even after monthly aggregation.

Debates. Mortality $\neq$ admissions. Warning-system behavioural responses may also differ. Still, the exposure engineering lesson transfers cleanly.

Implementation. Extend monthly climate outputs with:

- `days_in_hot_night_spell_ge5`
- `days_in_very_hot_spell_ge5`
- `longest_cold_run`
- a monthly indicator for whether a 2D3N-type event touches the month

These are already partially scaffolded; they should be first-class alternative exposures in modelling scripts, not orphan columns.

4 Pollution, influenza, and the cold pathway

4.1 Guo et al. (2025) — temperature effects amplified on polluted days

Adds. Over 2000–2019, PM_{2.5} amplified temperature–hospitalization associations in Hong Kong; NO₂ intensified circulatory risks; risks were higher on days with one or more elevated pollutants than on unpolluted days (Guo et al., 2025).

Inspires. Pollution is not only a confounder checklist item. It can modify temperature effects. Staged models should include interaction or stratified “polluted month” specifications once real EPIC series exist.

Debates. Ozone remains ambiguous: correlated with hot sunny weather, possibly pathway rather than pure confounder. Adjusting for O₃ in the primary heat model may attenuate the very association of interest.

Implementation. Preserve staged pollution modelling: (1) no pollutant; (2) single-pollutant (NO₂, PM_{2.5}, O₃ separately); (3) multi-pollutant sensitivity; (4) optional temperature $\times$ pollution product terms. Do not put O₃ into the default primary heat model without reporting the unadjusted heat estimate beside it. Consider an Ox-style redox summary ($\frac{1}{3}\text{NO}_2 + \frac{2}{3}\text{O}_3$) as an exploratory combined oxidant metric once real pollution data arrive (Yang et al., 2025).

4.2 Yang/Wei/Chong et al. (2025) — cold, influenza, and stroke

Adds. In a 22-year weekly Hong Kong analysis (~ 1.17 million stroke-related admissions), cold temperature and influenza activity (ILI+ rates) were both associated with higher stroke admissions (Yang et al., 2025). Absolute humidity was preferred over relative humidity as a humidity metric less entangled with temperature.

Inspires. Influenza is a scientifically serious co-exposure for cold-season stroke, not a decorative covariate. Absolute humidity is a better humidity construct than RH for some infection/cold pathways.

Debates. Weekly pooled stroke (ICD-9 430–434, primary and secondary diagnoses) is not our monthly principal-diagnosis AMI/IS/HS design. Their finding strengthens cold + flu hypotheses but does not license pooled-stroke primary analysis.

Implementation.

- Keep influenza in sensitivity models initially (assumption A23), with a clear path to promote it if CHP series are strong.
- Build absolute humidity from temperature and RH (Tetens vapour pressure) in the weather pipeline; do not rely on RH alone.
- Request strain-aware ILI+ if available; otherwise a total ILI+ or consultation proxy is still better than nothing.

5 Warm-climate winter excess and the “heat replaced cold” myth

5.1 Sipilä et al. (2017) and Lorking et al. (2020)

Adds. Stroke seasonality shows autumn/winter excess even in temperate Finland (Sipilä et al., 2017) and winter excess in tropical Thailand (Lorking et al., 2020). Warm climates are not heat-only climates for cerebrovascular burden.

Inspires. Our framing should be coexistence, not succession: emerging heat burden *and* persistent cold risk in an aging subtropical city.

Debates. Seasonality studies are not temperature-extreme studies. They support motivation, not exposure definitions.

Implementation. Retain cold days as co-primary. In the research note and Introduction, replace any “heat replacing cold” language with coexistence language. The validated HKO figure already shows cold days persisting through high-heat years (e.g., 13 cold days in 2021 alongside 61 hot nights).

6 Outcome validity: why principal diagnosis is non-negotiable

6.1 Cho et al. (2024) and Liu et al. (2025)

Adds. East Asian administrative-data validation work shows that carefully defined AMI and stroke algorithms can achieve high positive predictive values for first events (around 90% in the Korean NHIS validation), with lower but still useful PPVs for recurrent events (Cho et al., 2024; Liu et al., 2025). Episode construction matters.

Inspires. Insist on principal diagnosis, episode/transfer rules, and first versus recurrent distinctions where feasible.

Debates. Hong Kong HA coding may be ICD-9-CM, ICD-10, or mixed across 2013–2023. External PPVs do not automatically transfer.

Implementation. Data request to Roro should treat principal diagnosis, admission type, transfer linkage, and coding system-by-year as hard requirements. Small-cell suppression must never be coded as zero. If only secondary-diagnosis-inclusive extracts are available, demote claims accordingly.

7 Methodological scaffolding: what good time-series practice demands

7.1 Bhaskaran et al. (2013) and Gasparrini DLNMs

Adds. Time-series environmental epidemiology requires explicit handling of seasonality, long-term trend, time-varying confounders, lagged associations, and diagnostics (Bhaskaran et al., 2013). DLNMs flexibly capture nonlinear exposure–lag–response surfaces when daily data exist (Gasparrini et al., 2010). Globally, cold still accounts for a large share of temperature-attributable mortality even as heat receives more public attention (Gasparrini et al., 2015). Morbidity associations are often weaker and more heterogeneous than mortality associations (Ye et al., 2012).

Inspires. Our monthly design is a deliberate trade-off: we gain alignment with official monthly extreme counts, demographic offsets, and hospital-burden interpretation; we lose daily lag resolution. That trade-off should be stated as a feature of the question, not apologized for as a defect.

Debates. If HA ever releases daily aggregates, a companion daily DLNM would be scientifically powerful—especially for testing whether Guo-style HNe effects appear for principal-diagnosis AMI/IS/HS. Until then, do not overclaim lag biology.

Implementation. Current NB models with month fixed effects, smooth time trend, COVID phases, and population $\times$ days offsets are consistent with Bhaskaran-style discipline at monthly resolution. Diagnostics (residual ACF/PACF, overdispersion checks) should remain mandatory before any real-data inference. Consider a future optional daily module rather than forcing DLNM syntax onto 132 monthly points.

8 Synthesis: what remains defensible for this project

After reading the above as a system rather than a pile of citations, the defensible contribution is:

Among Hong Kong residents aged 35+, estimate **monthly** associations of **officially defined** thermal extremes (especially hot nights and cold days) with **principal-diagnosis inpatient** AMI, ischemic stroke, and hemorrhagic stroke during **2013–2023**, using age–sex population offsets, separate older age bands, COVID-period handling, and staged pollution adjustment—without assuming that heat has replaced cold, and without claiming primacy over recent daily hot-night hospitalization research.

8.1 Prespecified hypotheses (recommended)

1. Cold days are positively associated with monthly AMI and ischemic-stroke admission rates.
2. Official hot-night counts may show weak or null associations (compatible with Guo 2024); spell/intensity alternatives may be more informative, especially in older age groups.
3. Associations differ by diagnosis (AMI vs IS vs HS).
4. Ozone adjustment may attenuate heat associations if ozone partly lies on the pathway; total and pathway-adjusted estimates should both be reported.
5. Influenza adjustment may attenuate cold–stroke associations in sensitivity models.

8.2 Claims to avoid

- “First Hong Kong temperature–CVD study.”
- “First Hong Kong hot-night hospitalization study.”
- “Little is known about hot nights and cardiovascular admissions in Hong Kong.”
- “We will demonstrate a regime shift from cold to heat.”
- Direct numeric comparison of monthly RRs with Goggins daily per-°C estimates.

9 From literature to pipeline: implementation backlog

The following items translate the review into repository work. Items marked **[done/partial]** already exist in some form; others are next steps.

Literature cue	Pipeline / design action	Where
Guo 2024 intensity vs binary	Keep official hot nights primary; add spell/intensity alternatives; write null-interpretation rule	<code>config.yml</code> ; Methods; script 11
Wang 2019 5HN / 2D3N	Export monthly spell-ge5 and 2D3N-touch indicators as modelling covariates	script 02; variable dictionary
Yang/Chong 2025 absolute humidity	Compute monthly mean absolute humidity from T and RH (Tetens)	<code>utils.R</code> ; script 02
Yang/Chong 2025 influenza	Load CHP ILI series; flu sensitivity model	<code>data_raw/chp_flu</code> ; scripts 06, 11
Guo 2025 pollution modification	Staged pollutant models + optional polluted-month interactions; cautious O_3	scripts 04, 11; assumption ledger A31
Goggins 2012 subtypes	Mandatory diagnosis-stratified fits; no pooled stroke primary	script 11
Lam 2018 diabetes	Request DM flag if feasible; else document limitation	Roro memo; ledger A27
Cho/Liu coding validity	Principal diagnosis + episode rules as hard extract requirements	Roro memo
Bhaskaran diagnostics	Residual ACF/PACF and dispersion checks before inference	script 12
Gasparrini daily DLNM	Optional future daily module if HA daily aggregates become available	new script later

9.1 Creative but disciplined extensions

Three ideas are worth considering without diluting the core paper:

1. **Burden decomposition.** Even before causal claims, report how much of year-to-year variation in extreme counts is concentrated in multi-day spells. This is descriptive epidemiology that links HKO operations to hospital planning.
2. **Care-seeking versus biology under COVID.** Use all-cause emergency admissions (if obtained) as a utilization tracer. Temperature associations that vanish only in 2020–2022 may reflect access disruption rather than biology.
3. **Policy metric audit.** Treat the project partly as an audit of whether official extreme-day definitions, designed for warnings, also track diagnosis-specific hospital burden at monthly scale. That framing remains novel even after Guo 2024, because Guo’s null for HNday28 was daily, hot-season, and not principal-diagnosis AMI/IS/HS with demographic offsets.

10 Search documentation and limitations of this review

- **Date:** 2026-07-10 (this pass).
- **Sources:** PubMed, PMC, publisher DOI pages (Lancet RH-WP, ACS ES&T, Springer, PLOS, STOTEN), Epidemiology and Health, prior project evidence matrix.
- **Priority:** Hong Kong temperature–CVD; hot nights; stroke seasonality; coding validity; time-series methods.
- **Not claimed:** Exhaustive systematic review with dual screening and PRISMA flow. Living evidence matrix: `literature/evidence_matrix.csv`.
- **Human follow-up:** complete author list for Liu et al. (2025) from PDF; confirm HA coding system before locking ICD algorithms.

Bottom line

The literature does not ask us to abandon hot nights. It asks us to stop treating the official binary count as self-justifying, to keep cold in the foreground, to separate stroke subtypes, to stage pollution carefully, and to interpret monthly results as hospital-burden evidence rather than daily-trigger evidence. Those constraints make the project more honest—and, if executed well, more useful.

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